

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 16

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Ecological footprint: the area of the Earth's surface (global hectares) needed to provide the resources used by a person / population / activity and to absorb the wastes produced • Biocapacity: the biological productivity of an area (forests, cropland, grazing, fisheries, built land) - its ability to supply resources and absorb wastes, expressed in global hectares	2
01.2	• A hectare of land / sea with the world-average biological productivity • Biocapacities of real areas vary widely so a standard unit allows footprints and biocapacities of different places to be compared	2
01.3	• $8.1 / 1.7 = 4.8$ (accept 4.7-4.8)	1
01.4	• Higher HDI is associated with a larger footprint (USA 8.1 vs Bangladesh 0.8), though the UK achieves the same HDI as the USA with about half the footprint • Global footprint (2.8) exceeds global biocapacity (1.7) so humanity uses resources faster than they regenerate and CO₂ accumulates (overshoot) • Earth Overshoot Day is the date each year when a year's worth of resources / waste absorption has been used up; it falls earlier each year (8 August in 2016)	3
01.5	<i>Any 2 from:</i> • Renewable energy and energy conservation (carbon footprint is the largest component) • Reduced meat consumption / food waste; local seasonal food • Circular economy: repair, re-use, recycling; longer product lives • Public transport, efficient buildings	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 379-380; AQA spec 3.6.4.2.	

Question	Marking guidance	Marks
02.1	• South America: high loss ($5.1-5.5 \text{ Mha yr}^{-1}$) falling by about half by 2020 • Africa: loss steady at about 3.4 then rising to 3.9 by 2020 - now the highest • Europe: negative values - forest area increasing (afforestation exceeds loss), though the gain is shrinking	3
02.2	• South America: clearance for commercial cash crops (sugar cane, soya) and cattle ranching for export; HEP reservoirs; roads improving access; mining • Africa: growing rural populations clearing land for subsistence farming; fuelwood and charcoal collection above regrowth (charcoal wastes much of the wood's energy); fodder cutting in dry areas • Both: hardwood timber (teak, mahogany) from clear-felled mixed forest often not replanted; urban expansion	4
02.3	• They create barriers fragmenting wildlife populations • Improved access makes it easier to remove timber and farm produce, so clearance spreads along them	2
02.4	• Conifers grow quickly so plantations are replanted after harvest and yield a saleable crop within decades; slow-growing hardwoods (teak, mahogany) take too long for commercial plantations so they are taken from natural forest that is clear-felled and not replanted	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 367-368; AQA spec 3.5.3.3.	
03.1	• SSSI: UK legislation (Wildlife and Countryside Act); best examples of UK flora, fauna, geological features; management plan agreed with owner who must notify Operations Likely to Damage (ploughing, drainage, pesticides, tree planting / removal) • SPA: EU Birds Directive; places of international importance for birds (with SACs under the Habitats Directive form the Natura 2000 network; may restrict activities in surrounding areas) • Ramsar: Ramsar Convention (intergovernmental agreement); internationally important wetlands, originally for waterfowl, now wetland conservation and wise use generally • MCZ: UK marine legislation (Marine and Coastal Access Act) under the W&CA framework; protects marine habitats / species by restricting activities such as fishing and dredging; NNR: best SSSIs with complete communities	4
03.2	• Legal status: restrictions on damaging activities within the area (and sometimes in surrounding areas, eg drainage) • Management agreements between owners and statutory bodies to maintain suitable conditions • But it does not protect against external influences such as air pollution or climate change; very few sites are state-owned so protection depends on owner cooperation	3
03.3	• Much of the UK's wildlife lives in habitats created by farming (hay meadows, hedgerows) which only survive if the farming continues • Managing land for wildlife costs money / reduces income so farmers need payments (points for beetle banks, hedgerows, buffer strips, wild bird seed, low-input grassland) • Benefits everyone so society pays through central funding; legislation cannot compel positive management	3

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 30, 38-39; AQA spec 3.1.2.3.2.	
04.1	• In-situ: conserving species in their natural habitat with the whole community; ex-situ: away from the natural habitat, eg zoos, botanic gardens, seed banks, captive breeding • Needed when conservation in the wild is impossible or insufficient to prevent extinction (habitat gone, population too small, poaching)	2
04.2	<i>Any 4 from:</i> • Is the wild population threatened? • Is there a genetically diverse captive population? • Is in-situ conservation failing? • Is keeping a captive population realistic (habitat size, food, relationships, cost)? • Is release likely to be successful now or in future (suitable habitat, threats removed)?	4
04.3	• Blue whale: habitat size far too large to provide; food (huge quantities of krill / plankton) cannot be supplied; cost • Large blue butterfly: complex inter-species relationship - larvae are taken into ant nests where they overwinter feeding on ant grubs; requires specific ant species and thyme food plant	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-34; AQA spec 3.1.2.3.3.	
05.1	• Extensive cereal 5.0; intensive cereal 3.0 • Intensive beef 0.1	2
05.2	• Intensive cereal is most productive per hectare (60 GJ) but less efficient (3.0) than extensive (5.0) • Each extra unit of input gives a smaller extra yield (law of diminishing returns) • Beef has very low productivity and efficiency because energy is lost at an extra trophic level	3
05.3	• Any input that increases productivity but requires the use of energy (usually fossil fuel) • Manufacture of nitrate fertiliser (Haber process) and pesticides; fuel for machinery; pumping irrigation water; grain drying; food processing and transport; embodied energy of machinery	3
05.4	• Because of diminishing returns, the first units of input on unfertilised land give the largest yield increases • Moving inputs from intensively farmed land (where extra input adds little) to extensively farmed land gives a greater total yield - though land, cost and distribution limit this	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 324-327; AQA spec 3.5.1.2.	
06.1	• Extensive: low inputs of energy, food and chemicals so low cost and low environmental impact; sustainable; but low yield per area and dependent on natural productivity • Intensive: control of all limiting factors gives very high yield and profit per area; but high inputs (fishmeal from wild fish, energy), disease and lice at high density, pollution from wastes, antibiotic / pesticide use, escapes • Judgement depends on species, market and site	4

Question	Marking guidance	Marks
06.2	- Adults with desirable characteristics (fast growth, shape, colour) stripped of eggs and milt which are mixed for fertilisation - Eggs raised in aerated freshwater tanks, dead / diseased eggs removed daily; fry fed fishmeal pellets and moved through tanks as they grow - After 12-18 months smolts transferred to seawater cages, fed pellets until harvest at 3-5 kg	3
06.3	- About 10% of the world's mangroves have been cleared for coastal shrimp lagoons (up to 100 ha), removing nurseries and coastal protection - Lagoons need water exchange to remove wastes and aeration to prevent deoxygenation as wastes decompose - Eyestalk ablation stimulates breeding; high densities increase disease so antibiotics / pesticides used	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-356; AQA spec 3.5.2.3.	
07.1	B - radio waves do not travel through seawater; acoustic or satellite tags transmit position	1
07.2	- Shows survival, dispersal and breeding of released individuals so success can be assessed and future releases improved - Radio / GPS tags or collars; leg rings / wing tags on birds; camera traps; DNA from droppings to confirm identity and diet	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 402-407; AQA spec 3.1.2.3.5-3.1.2.3.6.	
08.1	- Trays of the same soil (texture, compaction, organic matter, depth) with different percentage vegetation cover (eg 0, 25, 50, 75, 100% grass), replicated - Same slope / gradient; simulated rain from a standard height at a standard rate and drop size (eg watering can / sprinkler) for a standard time - Collect the soil splashed / washed off (eg onto a sheet or into a trough at the base), dry and weigh it - Repeat each treatment several times and calculate means (and standard deviation) - Plot soil loss against cover; statistical test (Spearman's rank / t-test) for significance	5
08.2	<i>Any 3 from:</i> - Rainfall intensity / raindrop size / height - Soil texture, compaction, organic matter, depth - Permeability of the material beneath - Gradient - Root binding	3
08.3	- All variables except the independent variable can be standardised in the laboratory; in the field uncontrollable variables (weather) must be measured instead - Laboratory conditions may not represent natural conditions so results may not apply directly to real ecosystems	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 419-420; AQA spec 3.7.1, 3.7.2.2.	

Question	Marking guidance	Marks
09.1	• Isolation makes colonisation difficult: only marine species, seabirds, plants with floating seeds or chance arrivals (tortoises on logs, wind-blown birds) reach them • Whole taxa (terrestrial mammals, amphibians) may be absent so few original colonists • Colonists evolve in isolation under different selection pressures to fill the empty niches, producing unique species (Galapagos, Madagascar lemurs, Hawaii) • Endemics have small ranges and populations so are vulnerable to extinction	4
09.2	• The birds evolved without mammal predators so nest on the ground / in burrows with no defence • Rats eat eggs and chicks; numbers fell to very low levels (45 and 12 pairs) • After eradication breeding success rose about 7-12 fold, confirming predation was the main limiting factor	3
09.3	• Introduced mosquitoes carry avian malaria to which endemic birds have no resistance; birds survive only above about 1500 m where it is too cold for malaria • Warming allows mosquitoes / malaria to spread to higher altitudes, shrinking the safe range	2
09.4	• Control of developments and visitor numbers (biosecurity, quarantine of imports); protected areas; captive breeding of endemics	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 64-65; AQA spec 3.1.2.3.4.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Resources: wood, fibres (cotton, wool, silk), oils, fuels (wood, biofuels), food; new food species (morama bean, perennial cereals, eland, bison) better adapted to local conditions • Biomimetics: vehicle design, architecture, structures, adhesion, materials, ultrasound • Medicines from plants and animals (examples); most species unstudied - argument for habitat conservation • Physiological research using wildlife species; wildlife as pest control agents (ladybirds, Encarsia, Cactoblastis, Bt) • Genetic resources: crop wild relatives, Vavilov centres, gene pools • Ecosystem services: atmospheric composition, hydrological and biogeochemical cycles, soil maintenance, pollination, seed dispersal, habitat provision • Ethical / aesthetic / recreational arguments; the unknown value of species not yet investigated (precautionary principle) • Evaluation: which arguments are most persuasive to decision makers; economic valuation vs intrinsic value; judgement	9
10.2	• The variety and abundance of life (species, genes, habitats) in an area / on Earth	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	

Question	Marking guidance	Marks
11.1	Agriculture involves the control of abiotic and biotic factors to maximise food production. Discuss how this control can be achieved and evaluate its environmental impacts. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Species selection: market, range of tolerance, controllability • Abiotic control: temperature, light, water, soil fertility / nutrients, aeration, salinity, pH, CO₂, topography, relief, wind - methods and examples • Biotic control: pest groups, cultural control, biological control (successes and failures), sterile males, pheromones, genetic resistance, pesticides (properties), antibiotics, pollinators, soil biota • Manipulation of food species: population density, monocultures, selective breeding, asexual reproduction, cross-breeding, AI / embryo transfer, GM; energetics - intensive vs extensive, energy subsidies, ratios, FCR, food chain losses • Impacts: habitat loss / change, reduced biodiversity, introduced species, pesticide and nutrient pollution, GHGs, hydrological changes, soil erosion; positive: plagioclimax habitats • Social, economic and political influences shaping methods • Sustainable strategies and evaluation; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 306-337; AQA spec 3.5.1.	25

Question	Marking guidance	Marks
11.2	Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Setting priorities: IUCN Red List criteria, EDGE, endemic, keystone, flagship species, population dispersal • Legislation and protocols: Wildlife and Countryside Act, designated areas (SSSI to MCZ), CITES Appendices, IWC, EU CFP, ITTO - successes and limits of enforcement • Captive breeding and release: selection, difficulties, techniques, release site criteria, hard / soft release, monitoring; seed banks • Habitat conservation: creation, design (area, shape, corridors, age structure, colonisation), abiotic and biotic management, plagioclimax maintenance, population control, reintroductions • Habitat case studies (woodland, rainforest, coral reefs, islands, mangroves, Antarctica) illustrating threats and conservation efforts • Monitoring and new technologies informing decisions • Evaluation: examples of success (Iberian lynx, Hawaiian goose, Scilly rats, Yellowstone) and failure (ITTO, tigers); cost, politics, climate change; judgement on the most effective combination Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-71; AQA spec 3.1.2.3.	25